

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)

CHAPTER 06: HOW FORCES AFFECT MOTION & LAWS OF MOTION

TIME ALLOWED: 60 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-G-06

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET G - FINAL MASTERY & GRAND CHALLENGE TEST):

1. The question paper comprises **25 High-Tier Mastery & Olympiad-Standard Multiple Choice Questions** of **1 mark each**.
2. Tests **In-Depth Conceptual Mastery, Multi-Step Numericals & Rigorous Scientific Deductions**: Recommended target time is **60 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **GRAND CHALLENGE TIER:** This paper features comprehensive multi-step numericals, advanced Olympiad/NTSE-aligned scenarios, and rigorous experimental synthesis designed to achieve absolute 100/100 mastery.

Q.1 A constant net force of 12 N acts on a body of mass 3 kg initially at rest for 4 seconds. What is the final kinetic energy acquired by the body?

- (A) 96 J (B) 192 J
(C) 384 J (D) 48 J

Q.2 A machine gun fires 20 bullets per second, each of mass 30 g with muzzle velocity 500 m/s . What average backward force must the gunner exert to hold the gun stationary?

- (A) 150 N (B) 300 N
(C) 600 N (D) 15 N

Q.3 A body of mass m strikes a rigid vertical wall perpendicularly with speed v and rebounds elastically with the same speed v . What is the magnitude of the impulse imparted to the wall?

- (A) 0 (B) mv
(C) $2mv$ (D) $\frac{1}{2}mv$

Q.4 A rocket of initial mass $M_0 = 1000 \text{ kg}$ expels exhaust gas at a constant velocity $u = 2000 \text{ m/s}$ relative to the rocket. What rate of fuel consumption (dm/dt) is required just to overcome Earth's gravity at liftoff ($g = 9.8 \text{ m/s}^2$)?

- (A) 4.9 kg/s (B) 9.8 kg/s
(C) 2.45 kg/s (D) 19.6 kg/s

Q.5 Two masses $m_1 = 4 \text{ kg}$ and $m_2 = 6 \text{ kg}$ connected by a light string over a frictionless pulley (Atwood machine) are released. What is their common acceleration ($g = 9.8 \text{ m/s}^2$)?

- (A) 1.96 m/s^2 (B) 9.8 m/s^2
(C) 4.9 m/s^2 (D) 0.98 m/s^2

Q.6 A man of mass 70 kg stands on a weighing scale inside an elevator accelerating downwards at 2.8 m/s^2 . What apparent weight is indicated by the scale ($g = 9.8 \text{ m/s}^2$)?

- (A) 686 N (B) 490 N
(C) 882 N (D) 196 N

Q.7 What apparent weight does the same man experience if the elevator cable snaps and the elevator falls freely in free fall?

- (A) 686 N (B) Zero (apparent weightlessness)
(C) 490 N (D) 1372 N

Q.8 A ball of mass 0.2 kg falling vertically from a height of 5 m rebounds to a height of 3.2 m after striking the floor. What is the impulse imparted by the floor ($g = 10 \text{ m/s}^2$)?

- (A) $1.8 \text{ N} \cdot \text{s}$ (B) $3.6 \text{ N} \cdot \text{s}$
(C) $0.4 \text{ N} \cdot \text{s}$ (D) $2.0 \text{ N} \cdot \text{s}$

Q.9 A block of mass 10 kg resting on a rough horizontal floor ($\mu = 0.4$) is pulled with a horizontal force of 30 N. What is the acceleration of the block ($g = 9.8 \text{ m/s}^2$)?

- (A) 3.0 m/s^2 (B) 0 m/s^2
(C) 0.92 m/s^2 (D) 1.5 m/s^2

Q.10 Two bodies of masses 1 kg and 4 kg have equal kinetic energies. What is the ratio of their linear momentums ($p_1 : p_2$)?

- (A) 1 : 4 (B) 1 : 2
(C) 2 : 1 (D) 1 : 16

Q.11 Two bodies of masses 1 kg and 4 kg have equal linear momentums. What is the ratio of their kinetic energies ($E_{k1} : E_{k2}$)?

- (A) 4 : 1 (B) 1 : 4
(C) 2 : 1 (D) 1 : 2

Q.12 A 1000 kg car travelling at 20 m/s collides with a stationary 2000 kg truck and locks together. What is their common speed after the collision?

- (A) 6.67 m/s (B) 10.0 m/s
(C) 5.0 m/s (D) 13.3 m/s

Q.13 What is the angle between action and reaction forces?

- (A) 0° (B) 90°
(C) 180° (D) 45°

Q.14 A water jet from a hose pipe of cross-section 10 cm^2 strikes a vertical wall normally at speed 15 m/s and drops dead. What force does the water exert on the wall (density of water $= 1000 \text{ kg/m}^3$)?

- (A) 150 N (B) 225 N
(C) 1500 N (D) 75 N

Q.15 A 2 kg stone is dropped from a balloon ascending at 5 m/s . At the instant of release, the stone's velocity is:

- (A) Zero (B) 5 m/s upward
(C) 5 m/s downward (D) 9.8 m/s downward

Q.16 What is the kinetic energy of an object of momentum p and mass m ?

- (A) $\frac{p^2}{2m}$ (B) $\frac{p}{2m}$
(C) $2mp$ (D) $\frac{1}{2}mp^2$

Q.17 An object of mass 5 kg moves in a horizontal circle of radius 2 m at a constant speed of 4 m/s . What is the net force acting on the object?

- (A) Zero (B) 40 N towards the center
(C) 20 N tangentially (D) 10 N outward

Q.18 If the earth suddenly shrinks to half its current radius with mass remaining constant, the value of acceleration due to gravity on its surface will become:

- (A) Half (B) Double
(C) 4 times (D) Same

Q.19 A cricket ball of mass 150 g moving at 20 m/s is caught by a fielder in 0.1 s . The average force exerted on the fielder's hands is:

- (A) 30 N (B) 300 N
(C) 3 N (D) 15 N

Q.20 Why does a swimming duck propel itself forward through water?

- (A) It pushes water backward with webbed feet, and water pushes forward with an equal reaction force (B) Water pulls the duck
(C) Gravity pushes it (D) Air pushes it

Q.21 If linear momentum of a body increases by 50%, what is the percentage increase in its kinetic energy?

- (A) 50% (B) 100%
(C) 125% (D) 225%

Q.22 A passenger sitting in an open railway carriage on a train moving with uniform velocity tosses a coin vertically upward. The coin lands:

- (A) Exactly in his hand (B) Behind him
(C) In front of him (D) To the side

Q.23 What happens if the same coin is tossed while the train is accelerating forward?

- (A) It falls behind his hand (B) It falls in his hand
(C) It falls in front of his hand (D) It floats mid-air

Q.24 Two identical blocks of mass 5 kg each are connected by a cord on a frictionless table. A force of 20 N pulls the leading block. What is the tension in the connecting cord?

- (A) 20 N (B) 10 N
(C) 5 N (D) Zero

Q.25 Why do tires of automobiles have deep grooved treads?

- (A) To look aesthetic (B) To maintain optimum road grip by providing channels for rainwater evacuation, preventing hydroplaning skids
(C) To decrease fuel consumption (D) To make the car lighter

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET G)

Q. No.	Correct	Rigorous Scientific Solution & Formula Derivation
Q.1	(C)	$a = F/m = 12/3 = 4 \text{ m/s}^2$. Final velocity $v = at = 4 \times 4 = 16 \text{ m/s}$. Kinetic energy $E_k = \frac{1}{2}mv^2 = 0.5 \times 3 \times (16)^2 = 1.5 \times 256 = 384 \text{ J}$.
Q.2	(B)	Force $F = \frac{\Delta p}{\Delta t} = n \times m \times v = 20 \times 0.030 \times 500 = 300 \text{ N}$.
Q.3	(C)	Impulse $= p_f - p_i = (-mv) - (+mv) = 2mv$.
Q.4	(A)	Thrust $F = u \frac{dm}{dt} = M_0 g \Rightarrow \frac{dm}{dt} = \frac{M_0 g}{u} = \frac{1000 \times 9.8}{2000} = 4.9 \text{ kg/s}$.
Q.5	(A)	$a = \frac{m_2 - m_1}{m_1 + m_2} g = \frac{6 - 4}{6 + 4} (9.8) = \frac{2}{10} (9.8) = 1.96 \text{ m/s}^2$.
Q.6	(B)	$N = m(g - a) = 70 \times (9.8 - 2.8) = 70 \times 7.0 = 490 \text{ N}$.
Q.7	(B)	In free fall, $a = g \Rightarrow N = m(g - g) = 0 \text{ N}$ (weightlessness).
Q.8	(B)	Impact velocity $v_1 = \sqrt{2gh_1} = \sqrt{2 \times 10 \times 5} = 10 \text{ m/s}$ downward. Rebound velocity $v_2 = \sqrt{2gh_2} = \sqrt{2 \times 10 \times 3.2} = 8 \text{ m/s}$ upward. Impulse $= m(v_2 - (-v_1)) = 0.2 \times (8 + 10) = 0.2 \times 18 = 3.6 \text{ N} \cdot \text{s}$.
Q.9	(B)	Limiting static friction $f_s = \mu N = 0.4 \times (10 \times 9.8) = 39.2 \text{ N}$. Since the applied force (30 N) is less than 39.2 N, the block does not move ($a = 0$).
Q.10	(B)	$p = \sqrt{2m E_k}$. For equal E_k : $\frac{p_1}{p_2} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$.
Q.11	(A)	$E_k = \frac{p^2}{2m}$. For equal p : $\frac{E_{k1}}{E_{k2}} = \frac{m_2}{m_1} = \frac{4}{1}$.
Q.12	(A)	$m_1 u_1 = (m_1 + m_2) V \Rightarrow 1000 \times 20 = (1000 + 2000) V \Rightarrow 20000 = 3000 V \Rightarrow V = 20/3 \approx 6.67 \text{ m/s}$.
Q.13	(C)	Action and reaction forces act in exactly opposite directions along the same line of action (180°).
Q.14	(B)	$F = \rho A v^2 = 1000 \times (10 \times 10^{-4}) \times (15 \text{ m/s})^2 = 1 \times 225 = 225 \text{ N}$.
Q.15	(B)	Due to inertia of motion, the released stone inherits the balloon's instantaneous velocity of 5 m/s upward.
Q.16	(A)	$E_k = \frac{1}{2}mv^2 = \frac{(mv)^2}{2m} = \frac{p^2}{2m}$.
Q.17	(B)	$F_c = \frac{m v^2}{r} = \frac{5 \times (4)^2}{2} = \frac{5 \times 16}{2} = 40 \text{ N}$ directed radially towards the center.
Q.18	(C)	$g = \frac{GM}{R^2}$. If $R \rightarrow R/2$, $g' = \frac{GM}{(R/2)^2} = 4g$.
Q.19	(A)	$F = \frac{\Delta p}{\Delta t} = \frac{0.150 \times 20 \text{ m/s}}{0.1 \text{ s}} = \frac{3.0}{0.1} = 30 \text{ N}$.
Q.20	(A)	Newton's third law: The backward force exerted by the feet on water produces an equal forward reaction push on the duck.
Q.21	(C)	$E_k \propto p^2$. If $p' = 1.5 p$, $E_{k'} = (1.5)^2 E_k = 2.25 E_k$. Percentage increase $= (2.25 - 1) \times 100 = 125\%$.
Q.22	(A)	Due to inertia of motion, the coin maintains the forward horizontal velocity of the train and returns to the thrower's hand.
Q.23	(A)	The train speeds up while the coin is in the air, leaving the coin to fall behind the passenger's hand.

Q. No.	Correct	Rigorous Scientific Solution & Formula Derivation
Q.24	(B)	System acceleration $a = \frac{20}{5 + 5} = 2 \text{ m/s}^2$. Tension accelerates the trailing 5 kg mass: $T = m a = 5 \times 2 = 10 \text{ N}$.
Q.25	(B)	Tread patterns channel water away from the contact patch, maintaining high static frictional grip on wet roads.